

Principal Component Analysis

Introduction :

⇒ What is Principal Component Analysis?
It is an unsupervised machine learning algorithm used for feature extraction and dimensionality reduction.

⇒ PCA doesn't delete information randomly.
It tries to keep the essence of the data.

⇒ PCA is feature extraction.

⇒ Why is PCA considered an unsupervised?
Bcz it relies exclusively on input features, ignores target labels, and optimizes variance.

⇒ Everything in PCA revolves around one single idea i.e., variance.

⇒ What is Variance?

- How much the data points are spread around the mean.

- Large spread → large variance

- Small spread → small variance

- Formula:

$$\text{Var}(x) = \frac{\sum (x_i - \mu)^2}{n}$$

where,

x_i = each value

μ = mean

n = no. of observations

- Variance is a mathematical measure of spread.

⇒ Why Variance? Why not mean? Mean tells us the center of the data. The variance tells us how the data is distributed around the center.

⇒ Why doesn't PCA use Mean Absolute Deviation (MAD)?

The MAD is,

$$\text{MAD} = \frac{\sum |x_i - \mu|}{n}$$

↑

it also measures spread, but still we use variance for two main reasons:

- (i) Variance is mathematically smoother.
- (ii) PCA relies on covariance matrices.

⇒ What are Principal Components? After rotation: The new axes are called Principal Components.

If there are originally:

• 2 feature → PCA creates 2 principal comps.

• n feature → PCA creates n " " "

⇒ How does PCA Reduce dimensions?

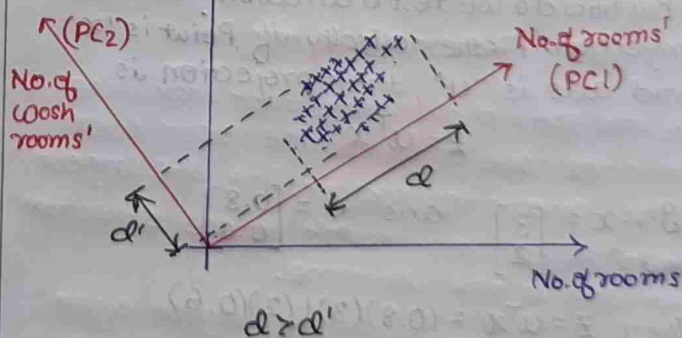
Suppose after rotation:

• PC1 explains 97% of the variance

• PC2 " 3% only

Then we may discard PC2. Instead of storing two features, we store only one. We lose only 3% of the info. but reduce the dimensionality by 50%.

⇒ No. of washrooms



⇒ Benefits of PCA:

- ① Faster Training
- ② Lower Memory Usage
- ③ Better visualization
- ④ Reduce Noise

⇒ The objective of PCA is: find the direction along which the variance of the projected data is maximum.

⇒ Maximising variance is bad when the target depends on low-variance features.

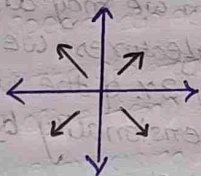
PCA : (Part 1)

⇒ PCA rotates the coordinate system until it finds the direction that preserves the maximum information.

⇒ Mathematics represent every direction using a unit vector. But why a unit vector? PCA cares only about the direction not how long the arrow is.

⇒ Suppose every possible unit vector is placed around the origin.

Each one represents one possible axis. PCA choose the best one.



⇒ But how do we test a direction? we use projection. Mathematically if Point is 'x' and axis is "u", then projection is

$$z = u^T x$$

e.g., $x = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$ axis $u = \begin{bmatrix} 0.8 \\ 0.6 \end{bmatrix}$

Then, $z = u^T x = (0.8)(3) + (2)(0.6) = 3.6$

∴ instead of storing (3, 2) we store 3.6. Dimension reduced.

⇒ How do we measure "Goodness" of a direction? using variance.

⇒ Which unit vector should PCA choose? PCA computes variance of every possible unit vector, then chooses the one with maximum variance. PCA solves with equation

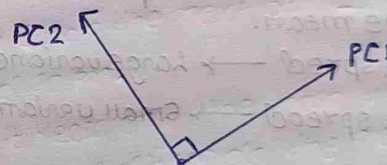
$$\max_{\|u\|=1} \text{Var}(u^T x)$$

⇒ But why maximum variance? To preserve the structure of the data.

⇒ Any special name for the unit vector that gives the maximum variance? First Principal component (PC1).

So, basically PC1 is the direction that preserves the greatest amount of info. from the original dataset after projection.

⇒ What about PC2? Later, after finding PC1, PCA searches for another direction that is $\perp$ (orthogonal) to PC1 and captures the maximum remaining variance. This becomes PC2.



⇒ But, there are ∞ many directions. Trying every direction one by one is not possible. So, we look forward for Covariance matrix.

⇒ Before it, what is covariance? Covariance measures how two variables change together. e.g.,

House	Rooms	Bathroom
A	2	1
B	3	2
C	4	3
D	5	4

Clearly, Room ↑
Bathroom ↑

∴ covariance is +ve

$\Rightarrow$ +ve covariance $\rightarrow$ both move together
 $-ve$ " $\rightarrow$ opposite movement
 0 " $\rightarrow$ No linear relationship
 $\therefore$ covariance tells direction of relationship, not strength.

$\Rightarrow$ Formula of covariance:

For feature X, Y ;

$$\text{Cov}(X, Y) = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

e.g.,

Rooms	Bathrooms	$\therefore \bar{x} = 3$ $\bar{y} = 2$				
2	1	Now, Rooms	Bathrooms	Deviation X	Deviation Y	Product
3	2					
4	3					
		2	1	-1	-1	1
		3	2	0	0	0
		4	3	1	1	1

$\therefore$ add products $\Rightarrow 1+0+1=2$

$$\therefore \text{covariance} = \frac{2}{n} = \frac{2}{3} \rightarrow +ve$$

$\Rightarrow$ Also, if we are finding covariance of a variable with itself then we get the variance.

$$\text{Cov}(X, X) = \frac{1}{n} \sum (x_i - \bar{x})^2 = \text{Var}(X)$$

$$\therefore \text{Var}(X) = \text{Cov}(X, X)$$

$\Rightarrow$ Then why covariance matrix? As no. of pairs increases, handling them separately be difficult. we arrange them in a matrix called covariance matrix.

e.g., for three features $\rightarrow x_1, x_2, x_3$

The cov matrix is,

$$S = \begin{bmatrix} \text{Cov}(x_1, x_1) & \text{Cov}(x_1, x_2) & \text{Cov}(x_1, x_3) \\ \text{Cov}(x_2, x_1) & \text{Cov}(x_2, x_2) & \text{Cov}(x_2, x_3) \\ \text{Cov}(x_3, x_1) & \text{Cov}(x_3, x_2) & \text{Cov}(x_3, x_3) \end{bmatrix}$$

$\Rightarrow$ Covariance matrix:

(i) For n features $(x_1, x_2, \dots, x_n)$

$$S = \begin{bmatrix} \text{Cov}(x_1, x_1) & \text{Cov}(x_1, x_2) & \dots & \text{Cov}(x_1, x_n) \\ \text{Cov}(x_2, x_1) & \text{Cov}(x_2, x_2) & \dots & \text{Cov}(x_2, x_n) \\ \vdots & \vdots & \ddots & \vdots \\ \text{Cov}(x_n, x_1) & \text{Cov}(x_n, x_2) & \dots & \text{Cov}(x_n, x_n) \end{bmatrix}$$

(ii) Special property,

$$\text{Cov}(A, B) = \text{Cov}(B, A)$$

$\therefore$ Cov. matrix is a sym. matrix,

(iii) what does each cell mean?

Let,

$$S = \begin{bmatrix} 5 & 2 \\ 2 & 7 \end{bmatrix}$$

$\therefore$ top-left,

$$\text{Var}(x_1) = 5$$

i.e., spread of feature 1.

$\therefore$ Bottom-right,

$$\text{Var}(x_2) = 7 \text{ i.e., spread of feature 2.}$$

$\therefore$ top-right

$$\text{Cov}(x_1, x_2) = 2 \text{ i.e., relationship b/w the two features}$$

(iv) Remarkable result:

$$\text{Var}(u^T x) = u^T S u$$

where, $u \rightarrow$ unit vector

$S \rightarrow$ covariance matrix

(v). Geometrically, it summarises the shape, spread and relationships within the data cloud.

(vi). PCA uses the covariance matrix bcz the variance of the data projected into any unit vector u can be written compactly

as,

$$\text{Var}(u^T x) = u^T S u$$

$\Rightarrow$ Is the equation $\text{Var}(u^T x) = u^T S u$ is enough? No, now we need to find out the unit vector u that maximises this expression.

$\Rightarrow$ But, how? We will use transformations. Since, most vectors are both rotated & scaled, we will need a vector whose dirⁿ does not change, only their length changes. These special vectors are called eigenvectors.

$\Rightarrow$ But, what is an eigenvector? It's a vector whose direction remains unchanged after a matrix transformation, only its length changes.

$\Rightarrow$ Is there any term specifically for change in the length? Yes, eigen values.

$\Rightarrow$ What are eigen values? It tells us how much the corresponding eigen vector is stretched or shrunk.

$\Rightarrow$ The fundamental equation:

$$A v = \lambda v$$

where,

$A \rightarrow$ covariance matrix

$v \rightarrow$ Eigenvector (a special direction)

$\lambda \rightarrow$ Eigenvalue (amount of stretching along that direction)

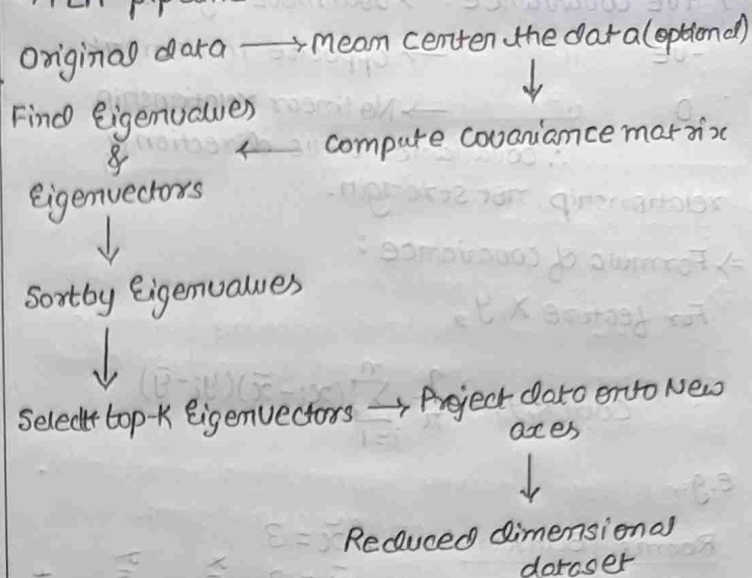
$A v \rightarrow$ Transform the vector using the matrix

$\lambda v \rightarrow$ Scale the vector

So, the optimal directions that maximise variance are precisely the eigenvectors of the covariance matrix. Also, the largest eigen values corresponds to the direction with the largest variance.

$\Rightarrow$ The eigenvector with the largest eigen value is chosen as PC1 bcz it maximizes the variance of the projected data.

$\Rightarrow$ PCA pipeline:



$\Rightarrow$ Why PCA works so well? Bcz real-world data often contains redundancy.

$\Rightarrow$ So, summarising:

- ① It looks at the shape of the data cloud.
- ② It finds the direction where the shadows (projection) is most spread out.
- ③ That direction becomes PC1.
- ④ It then finds the next best $\perp^{\circ}$ direction (PC2), and so on.
- ⑤ Finally, it keeps only the most informative directions and discards the rest.